

Independent replication of Griffiths & Niu (1996) — Semiclassical Fourier Transform for Quantum Computation

arXiv:quant-ph/9511007 · PRL **76**, 3228

Ollie (OpenClaw QC-200 wave, subagent report)

2026-07-05

Abstract

We independently replicate the central quantitative claim of Griffiths & Niu (1996): the quantum Fourier transform (QFT) followed by measurement can be re-implemented as a strictly *semiclassical* circuit — Hadamards + single-bit measurements + single-bit phase gates classically conditioned on prior measurement outcomes — producing *identical* measurement statistics as the standard QFT circuit that uses coherent controlled-phase two-qubit gates. We build both variants directly in Qiskit (v2.5.0) using the modern dynamic-circuits `if_test` primitive for the classical feed-forward, execute them on the Aer state-vector simulator, and verify equivalence over *all* computational-basis inputs at $n \in \{3, 4\}$ qubits and over a family of Shor-style periodic superposition inputs at $n \in \{3, 4\}$. The maximum total-variation distance (TVD) between the standard-QFT and the semiclassical-QFT measurement distributions was 0.032 (on $|x\rangle$ inputs, 8192 shots per input) and 0.011 (on periodic superpositions, 16384 shots), both well within shot noise. Gate counts match the paper exactly: the semiclassical variant uses 0 two-qubit gates versus $n(n-1)/2$ for the standard variant. **Verdict: REPLICATED.**

1 Paper summary

Shor’s algorithms (factoring, discrete log) end with a QFT on an $n = s+1$ -qubit register followed by a computational-basis measurement. The standard QFT circuit ([?] Fig. 1) uses n Hadamards and $n(n-1)/2$ two-qubit controlled-phase gates:

$$|a\rangle \xrightarrow{\text{QFT}} F|a\rangle = \frac{1}{\sqrt{q}} \sum_{c=0}^{q-1} e^{2\pi i ac/q} |c\rangle, \quad q = 2^n.$$

Griffiths & Niu observe that because controlled-phase gates leave their control bit unchanged, one may equally well *measure* the control immediately after its Hadamard and then have the classical outcome deterministically decide whether or not to apply a single-qubit phase to each less-significant qubit. Iterating this replacement throughout the circuit yields their Fig. 2: a bit-by-bit sequential protocol in which the j -th qubit is (a) Hadamard-ed, (b) measured into a classical bit c_j , then (c) each not-yet-measured qubit $k < j$ has a phase gate $P(2\pi/2^{j-k+1})$ applied to it *only if the measurement result c_j was 1*. Zero coherent two-qubit gates survive.

The paper’s central quantitative claim (its ”success in simplifying”) is that this semiclassical circuit produces *identical* measurement statistics as the standard QFT circuit followed by measurement. It also makes a precise gate-count claim, and offers consistent-histories intuition for why the substitution is valid.

2 Claims table

ID	Claim	Testable?	Tested here?
C1	Semiclassical QFT (Fig. 2) reproduces the measurement distribution of the standard QFT (Fig. 1).	Yes	Yes
C2	Semiclassical QFT uses 0 two-qubit gates and $n(n-1)/2$ single-qubit phase gates (classically conditioned).	Yes	Yes
C3	Standard QFT uses $n(n-1)/2$ two-qubit controlled-phase gates.	Yes	Yes
C4	The classical phase-update rule is $\phi' = \phi/2 + c/4$ from Eq. (11); phase feed-forward is between consecutive boxes.	Yes	Yes (implicit in per-qubit P correction u
C5	The substitution is justified by a consistent-histories argument.	Interpretive	No (mathematical, not empirical)

3 Method (exact commands and versions)

Environment.

- Host: CherryRd (macOS Darwin 25.3.0), CPU only.
- Python: 3.13 inside a project-local venv.
- Qiskit 2.5.0, qiskit-aer 0.17.2, NumPy latest, no GPU.
- LLM inference: none required (this is a math/simulation replication); per QC-100 wave rule we only use free endpoints (Argo localhost:44497), and in fact used none here.

Files.

- `report/evidence/replicate_semiclassical_qft.py` — builds both circuit variants, runs Aer over all $|x\rangle$ inputs at $n=3,4$, computes TVDs against the analytic QFT distribution and against each other, writes `results.json` and `summary.txt`.
- `report/evidence/replicate_periodic_input.py` — replaces the $|x\rangle$ input with a Shor-style periodic superposition $|\psi_p\rangle = \frac{1}{\sqrt{K}} \sum_{k=0}^{K-1} |k \cdot p \bmod 2^n\rangle$ for several (n,p) , so the target QFT distribution is highly non-uniform (concentrated on c with $c \cdot p/2^n \in \mathbb{Z}$). Writes `results_periodic.json`.

Standard-QFT circuit (built from primitives, no library QFT). For each j from $n-1$ down to 0: `H(q[j])`, then for k from $j-1$ down to 0: `cp(2*pi/2^(j-k+1), q[k], q[j])`. After the loop, measure `q[j] -> c[n-1-j]` so that the classical register read little-endian equals the integer c used in the paper.

Semiclassical circuit (dynamic circuit). For each j from $n-1$ down to 0: `H(q[j])`, then `measure(q[j], c[n-1-j])`, then for each $k < j$: with `qc.if_test((c[n-1-j], 1))`: `qc.p(2*pi/2^(j-k+1), q[k])`. The Qiskit `if_test` block is exactly the classical-feed-forward Griffiths & Niu describe as “a pulse of several volts traveling down a coaxial cable”. Aer’s `if_test` handling is well documented and was cross-verified against Qiskit’s built-in dynamic-circuit primitives.

Reference distribution. For $|x\rangle$ inputs the analytic post-QFT distribution is uniform $1/2^n$ (this is one motivation for the periodic-input test, since the $|x\rangle$ probability can hide phase bugs). For periodic-superposition inputs we compute the exact QFT-probability distribution $|(M|\psi_p\rangle)_c|^2$ where $M_{c,x} = e^{2\pi i x c/N} / \sqrt{N}$ in NumPy.

Shots & seeds. shots=8192 per input for the $|x\rangle$ sweep; shots=16384 per input for the periodic-superposition sweep; Aer seed derived from a fixed integer + input index for reproducibility.

4 Results vs paper

4.1 Computational-basis inputs $|x\rangle$, $x = 0..2^n - 1$

n	inputs	shots	max TVD (full vs analytic)	max TVD (semi vs analytic)	max TVD (full vs semi)
3	8	8192	0.0161	0.0131	0.0195
4	16	8192	0.0201	0.0186	0.0316

Both the standard QFT and the semiclassical QFT match the analytic uniform distribution to within Monte-Carlo shot noise, and match each other to within the same tolerance. Every one of the $8 + 16 = 24$ per-input pairwise TVDs is below the pre-registered 0.08 threshold (chosen at $\sim 4\sigma$ of the binomial noise for 16 bins/8192 shots), and the maximum observed is 0.0316.

4.2 Shor-style periodic superposition inputs

Here the analytic post-QFT distribution is a comb concentrated on the divisor-lattice values.

n	period p	expected peaks	observed peaks	shots	TVD(full—ana)	TVD(semi—ana)	TVD(full—semi)
3	2	{0,4}	{0,4}	16384	0.0018	0.0011	0.0029
3	4	{0,2,4,6}	{0,2,4,6}	16384	0.0035	0.0045	0.0063
4	2	{0,8}	{0,8}	16384	0.0018	0.0012	0.0006
4	4	{0,4,8,12}	{0,4,8,12}	16384	0.0035	0.0074	0.0067
4	8	{0,2,4,6,8,10,12,14}	{0,2,4,6,8,10,12,14}	16384	0.0060	0.0089	0.0107

The peak structure is reproduced exactly by both circuit variants in all 5 cases, and the empirical TVDs between the standard and semiclassical outputs are ≤ 0.011 — again well within shot noise for 16384 shots on ≤ 16 bins. This is the strong test: the periodic-superposition QFT distribution is far from uniform, so any phase-management bug in the feed-forward would be visible immediately.

4.3 Gate counts

n	Standard-QFT 2q CP gates	Semiclassical 2q CP gates	Semiclassical 1q classically-cond. phase gates
3	3	0	3
4	6	0	6
5	10	0	10
8	28	0	28

Exactly the $n(n-1)/2$ scaling Griffiths & Niu quote. Note the number of classically-conditioned 1-qubit phase gates matches the number of eliminated 2-qubit gates one-for-one; this is a consequence of unrolling the paper’s compact “per box” recipe (Eqs. 10–11) into the pairwise pattern used in Fig. 1.

5 Verdict

REPLICATED. All quantitative claims of Griffiths & Niu are reproduced within Monte-Carlo error using real Qiskit dynamic circuits: (i) semiclassical output distribution = standard-QFT output distribution on both trivial basis and non-trivial periodic inputs, (ii) gate-count scaling as claimed, (iii) zero coherent two-qubit gates in the semiclassical variant. The consistent-histories argument (C5) is an interpretive framing rather than a testable empirical claim, so we do not attempt to verify it; the empirical equivalence which the interpretation is meant to explain *is* verified.

6 Open Questions

- Q1. Latency-vs-fidelity crossover on real dynamic-circuit hardware.** Griffiths-Niu’s Fig. 2 assumes ideal projective measurements with zero delay before the classical signal reaches the next box. On real hardware, mid-circuit measurement latency is $\sim 1\mu\text{s}$ and the remaining qubits decohere during that window. What is the threshold ratio $T_2/(t_{\text{meas}} + t_{\text{classical_routing}})$ at which the semiclassical QFT begins to *lose* fidelity relative to the standard QFT, when we insert per-qubit depolarizing/ T_2 dephasing during the classical-feed-forward window? Our Aer replication used an ideal zero-latency `if_test` block, which trivially matches the standard QFT — the paper predates real dynamic-circuit hardware and so cannot answer this.
- Q2. Shot-noise scaling of TVD(full|semi) with number of Fourier peaks.** In our periodic-input tests ($n = 4, p = 8$) the semiclassical result showed the largest full-vs-semi TVD (0.011) of any case — roughly $3\times$ larger than the $p = 2$ case at the same shot budget. Is this genuinely extra shot-noise correlation between measurement outcomes and later phases (in which case the semiclassical variant has *strictly higher* shot-noise variance than the standard variant), or a transpiler artifact of the unrolled `if_test` blocks? Formalize: prove or disprove that the joint distributions of the n classical outcomes are *equal* as measures (the paper proves only marginals).
- Q3. Partially-semiclassical QFT for non-terminal QFT sub-blocks.** Griffiths-Niu eliminate every two-qubit gate only because the QFT is followed immediately by full measurement. In algorithms that use the post-QFT state coherently (amplitude estimation without terminal measurement, QSVT sub-blocks), the substitution does not apply as stated. Is there a hybrid variant that measures only the first $m < n$ qubits semiclassically — converting $m(n-m)$ two-qubit gates into classically-conditioned single-qubit gates, leaving the remaining coherent register live? What is the Pareto frontier of (2q-gate count, state fidelity to ideal QFT on the $n-m$ remaining qubits) as m varies?
- Q4. A channel-identity proof of equivalence, without consistent histories.** The paper’s argument uses the consistent-histories formalism. Can the equivalence be written directly as a CPTP-channel identity of the form (measure then classical-controlled P) = (CP then measure) on the two-qubit reduction, verified element-wise in Kraus form, then

inducted across the QFT? A non-specialist-friendly channel-identity appendix would make the result portable to circuit-QED / dynamic-circuit engineers unfamiliar with consistent-histories notation.

Q5. Deferred-measurement premium on nearest-neighbour connectivity. On hardware without `if_test` support, the semiclassical variant must be recompiled via the principle-of-deferred-measurement — which reintroduces two-qubit gates. Does the deferred-semiclassical circuit ever yield a strictly shorter total depth than the original standard QFT after transpilation onto a linear-nearest-neighbour coupling map at $n=10..20$? If yes, quantify the dynamic-circuits premium; if no, formalize this as evidence that the entire Griffiths-Niu benefit is contingent on hardware-level classical feed-forward support.

References

- [1] R. B. Griffiths and C.-S. Niu, “Semiclassical Fourier Transform for Quantum Computation”, Phys. Rev. Lett. **76**, 3228 (1996); arXiv:quant-ph/9511007.